Ratios

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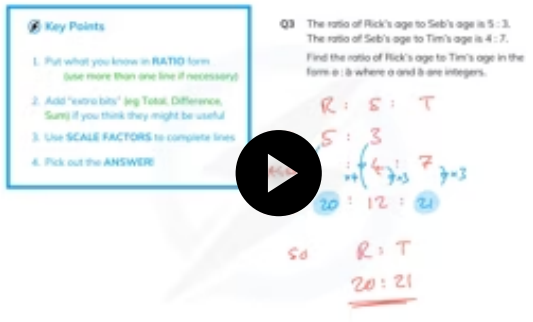

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Ratios

What is a ratio?

- A **ratio** is a way of comparing one **part** of a **whole** to another
  - **Ratios** are used to compare **one part** to **another part**

What do ratios look like?

- Ratios involve **two or three different numbers** separated using a **colon**
  - E.g. 2 : 5, 3 : 1, 4 : 2 : 3
- In all ratio questions, who or what is **mentioned first** in the question, will be associated with the **first part of the ratio**
  - E.g. The cake recipe with flour and butter in the ratio 2 : 1
    - 'Flour' is associated with '2' and 'butter' is associated with '1'
- The numbers in a ratio tell us, for **each quantity** involved, its **proportion of the whole**
  - In the ratio **4 : 3**
    - The **first quantity** comprises 4 parts (of the whole)
    - The **second quantity** comprises 3 parts (of the whole)
    - In total, the **whole** is made up of 4 + 3 = 7 parts

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it

- The **second quantity** comprises 5 parts (of the whole)
- The **third quantity** comprises 3 parts (of the whole)
- In total, the **whole** is made up of 2 + 5 + 3 = 10 parts

How are ratios and fractions linked?

Help me understand

- Fractions and ratios are closely linked and are used to compare a **part of a whole**
  - A **fraction** compares **a part to the whole**
  - A **ratio** compares **one part to another part**
- For example, a pizza is sliced into **8 pieces**, and shared between **two people** such that the **first person receives 5 slices**, and the **second person receives 3 slices**
  - As a **fraction** (of the whole)
    - The first person receives $\frac{5}{8}$ of the pizza
    - The second person receives $\frac{3}{8}$ of the pizza
- The **ratio** of slices of the first person to the second person is 5 : 3
  - The 8 does not appear in the ratio but is obtained by adding the parts together (5 + 3 = 8)
- Unlike ratios, **fractions** can be converted/expressed as **percentages** or **decimals**

Worked Example

A pot of money is shared between three friends, Dave, John and Mary.
Dave receives $450, John receives $200 and Mary receives $350.

(a) Find the total amount of money in the pot.

Answer:

Add up the three separate amounts

$$450 + 200 + 350 = 1000$$

$1000

(b) Write down the ratio of money received by Dave, John and Mary.
(There is no need to simplify the ratio.)

Answer:

Be careful with the order
Dave gets mentioned first, so 450 will be the first part of the ratio, then John and finally Mary

450 : 200 : 350

Answer:

Fractions are compared to the whole, so this will be 'Mary's money' "out of" 'total money'

$$\frac{350}{1000}$$

What is an equivalent ratio?

- **Equivalent ratios** are two ratios that represent the **same proportion** of quantities within a whole
  - E.g. The ratio **5 : 10** is equivalent to **20 : 40**
- Equivalent ratios are frequently used when the values involved take on a **real-life meaning**
  - E.g. A cake recipe involves flour and butter being mixed in the ratio 3 : 2
    - 3 g of flour and 2 g of butter would not lead to a very big cake
    - An equivalent ratio of 300 : 200 gives a more realistic 300 g of flour and 200 g of butter

How do I find an equivalent ratio?

- You can find an **equivalent ratio** by **multiplying** (or **dividing**) **each part** of the ratio by the **same value**
  - E.g. Multiply each part of the ratio 2 : 3 : 7 by 4 to find an equivalent ratio of 8 : 12 : 28
  - Ratios can be scaled up or down to suit the context of a question
- The size of each part in the ratio, **relative** to the others, is still the same
  - The **actual values** in the equivalent ratio may be **more meaningful** in the context of the situation
- Finding equivalent ratios is **similar** to finding **equivalent fractions**
  - However it is crucial to remember that 1 : 4 is **not** equivalent to $\frac{1}{4}$

Examiner Tips and Tricks

Writing down what you are **doing** to each part of the ratio helps show your working and makes it easier to keep track of what you are doing.

E.g.

$$15 : 20$$

Worked Example

The ratio of cabbage leaves eaten by two rabbits, Alfred and Bob, is 7 : 5.

(a) Write down an equivalent ratio that would involve a total of 48 cabbage leaves being eaten.

Answer:

We have information about the whole so first add up the parts of the ratio to find how many parts make the whole

$$7 + 5 = 12$$

We require a total of 48 so divide this by 12 to find the multiplier

$$48 \div 12 = 4$$

Answer the question by multiplying each part of the ratio by this multiplier

$$\begin{array}{ccc} & A : B & \\ & 7 : 5 & \\ \times 4 \downarrow & & \downarrow \times 4 \\ & 28 : 20 & \end{array}$$

28 : 20

(b) On another occasion Bob eats 35 leaves.

Find out how many leaves Alfred eats.

Answer:

Use the information about Bob to find the multiplier using division

$$35 \div 5 = 7$$

Now multiply Alfred's part of the ratio by the multiplier to answer the question

$$\begin{array}{ccc} & A : B & \\ & 7 : 5 & \\ \times 7 \downarrow & & \downarrow \times 7 \\ & 49 : 35 & \end{array}$$

Alfred eats 49 cabbage leaves

What is a simplified ratio?

- A ratio is in its **simplest form** when
  - All of the values in the ratio are **integers**
  - There are no **common factors** between each of the values in the ratio
  - E.g. The simplest form of the ratio **45 : 30** is **3 : 2**

How do I simplify a ratio?

- **Divide each part** of the ratio by the **same value**
  - This value should be a **common factor** of all parts of the ratio
    - Ideally, the **highest common factor** **(HCF)** should be used to get the ratio into its simplest form in one go
    - If the HCF is not used, we can repeat the process of simplifying
  - E.g. Divide all parts of the ratio **30 : 66 : 12** by 6 to find the ratio in its simplest form **5 : 11 : 2**

Worked Example

Amber and Naomi are sharing a large cake that has been cut into 48 pieces.
Amber receives 30 of these pieces, Naomi receives the rest.

Write down the ratio, in its simplest form, of the number of pieces of cake that Amber receives to the number of pieces of cake that Naomi receives.

Answer:

First use subtraction to find the number of pieces that Naomi receives

$$48 - 30 = 18$$

Next write down the ratio without simplifying
Make sure you have the order correct ('Amber to Naomi')

$$30 : 18$$

Look for a common factor of both 30 and 18 to simplify the fraction
6 is the highest common factor
Divide both parts of the fraction by 6

$$\begin{array}{ccc} A & : & N \\ 30 & : & 18 \\ \div 6 \downarrow & & \downarrow \div 6 \\ 5 & : & 3 \end{array}$$

5 : 3

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